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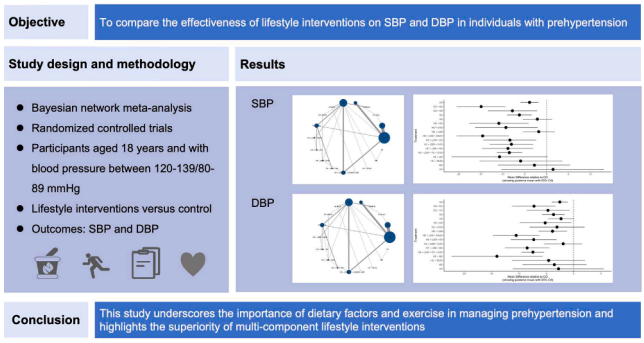
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Comparative effectiveness of lifestyle interventions for prehypertension: A network meta-analysis of randomized controlled trials

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GRAPHICAL ABSTRACT



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ABSTRACT

This study aims to compare the effectiveness of lifestyle interventions on systolic blood pressure (SBP) and diastolic blood pressure (DBP) in individuals with prehypertension. A systematic literature search was conducted in PubMed, Web of Science, Embase, Cumulative Index to Nursing and Allied Health Literature, China National Knowledge Infrastructure, and the Wanfang database from inception to December 2023. The analysis included randomized controlled trials with either parallel or cross-over designs. Utilizing a Bayesian network meta-analysis approach, we included 67 studies evaluating 17 lifestyle interventions compared with control. For

Abbreviations: ACC/AHA, American college of cardiology/American heart association; CINAHL, Cumulative Index to Nursing and Allied Health Literature; CNKI, China National Knowledge Infrastructure; CO, control; CrI, credible interval; DASH, dietary approaches to stop hypertension; DIC, deviance information criterion; DBP, diastolic blood pressure; DS, dietary supplement; EX, exercise; GHD, general healthy diet; GBD, global burden of disease; HE, health education; ID, identifier; I^2, I-square statistic; ISH, the International Society of Hypertension; LMICs, low- and middle-income countries; LSM, lifestyle modification; MA, massage; MD, mean difference; ME, meditation; PC, psychological care; PICOS, participant, intervention, control, outcomes, and study design; PRISMA, the preferred reporting items for systematic reviews and meta-analyses; ROB, risk of bias; RT, relaxation therapy; SBP, systolic blood pressure; SD, standard deviation; SUCRA, surface under the cumulative ranking curve; TEAS, transcutaneous electrical stimulation of acupuncture points; UK, the United Kingdom; USA, the United States of America; WHO, the World Health Organization..

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SBP, dietary supplement plus exercise was the most effective lifestyle intervention, with a mean difference (MD) of -14.86 (95 % credible interval [CrI] -20.44 to -9.35). For DBP, health education plus meditation was the most effective lifestyle intervention (MD -13.99, 95 % CrI -22.25 to -5.65). Subgroup analyses showed that the effectiveness of the interventions varied by duration, age, and economic level of the region. This study underlines the importance of dietary factors and exercise in the management of prehypertension and highlights the superiority of multi-component lifestyle interventions. However, the effectiveness of these interventions varies based on participant characteristics and intervention type, emphasizing the need for tailored, population-specific approaches.

1. Introduction

Hypertension is the primary cause of cardiovascular disease and premature death worldwide, affecting 33 % of adults aged 30–79 years, according to age-standardized estimates from the World Health Organization (WHO) [1]. Despite the stable global age-standardized prevalence, the hypertensive population has doubled from 1990 to 2019, from 650 million in 1990 to 1.3 billion in 2019 [2]. This increase is mainly due to the rapid ageing of the population and the epidemiological transition, particularly in low- and middle-income countries (LMICs), where the prevalence of hypertension is higher (31.5 %, 1.04 billion people) than in high-income countries (28.5 %, 349 million people) [3].

Considering the economics and public health burden of elevated blood pressure, both the WHO and the International Society of Hypertension (ISH) advocate for a comprehensive approach to the prevention and treatment of hypertension [4,5]. Lifestyle risk factors, such as unhealthy diet, sedentary physical activity level, overweight and obesity, smoking, and poor sleep quality, are significantly associated with the development of hypertension [6–9]. While pharmacological treatment is effective in controlling blood pressure, lifestyle recommendations have served as a first-line strategy for blood pressure management [10,11].

Prehypertension, also known as high-normal hypertension, elevated blood pressure, or borderline hypertension, refers to blood pressure levels between normal and hypertensive levels[12]. The term "prehypertension" was defined in 2003 as blood pressure in the range of 120–139 mmHg (systolic) or 80–89 mmHg (diastolic) [13]. As hypertension guidelines have evolved over the years, they have increasingly included a broader at-risk population by adopting different blood pressure cut-offs [14,15]. This evolution is supported by epidemiological evidence indicating that individuals with prehypertension are at an increased risk of developing hypertension and cardiovascular disease [9,12,16–18]. Up to half of the morbidity associated with blood pressure occurs in individuals with a SBP $<$140 mm Hg [19]. Moreover, research suggests that prehypertension represents a distinct phenotype in the progression from health to clinically overt disease, characterized by significant pathophysiological, structural, and functional hemodynamic changes [20]. Given these concerns, lifestyle interventions have shown promising efficacy in helping individuals with prehypertension achieve normal blood pressure levels [21].

To address the hypertension challenge, it is essential to identify effective lifestyle measures for prehypertension through a systematic evaluation of intervention evidence. While many studies have been conducted on this topic, several gaps have been identified. Firstly, several previous meta-analyses have included both prehypertensive and hypertensive populations, or have included both pharmacology and non-pharmacology interventions, making it difficult to specifically identify effective lifestyle interventions for prehypertension [22–25]. Secondly, the broad scope of "lifestyle" requires a comparative study of different non-pharmacological methods to clarify effects and research gaps [26,27]. Thirdly, the heterogeneity of treatment effects of lifestyle interventions highlights the need to consider intervention duration and socioeconomic factors when assessing the effectiveness of interventions [28,29].

This network meta-analysis aims to fill existing gaps by comprehensively evaluating the effectiveness of non-pharmacological lifestyle interventions for individuals with prehypertension. Additionally, subgroup analyses will explore the effect of intervention duration, age, and economic level of the region on effectiveness. The findings aim to inform evidence-based lifestyle recommendations for the prehypertensive population.

2. Methods

2.1. Literature search process

The literature search was conducted using multiple databases, including PubMed, Web of Science, Embase, Cumulative Index to Nursing and Allied Health Literature (CINAHL), China National Knowledge Infrastructure (CNKI), and Wanfang. The search period was from inception to December 2023. The search strategy included a comprehensive range of terms related to "prehypertension" or its symptoms, "lifestyle" or specific lifestyle factors, "trial," and "adults", the literature search strategy is shown in Table S1. References were screened, organized, and managed using EndNote 20. The study protocol was registered on the International Prospective Register of Systematic Reviews (PROSPERO; CRD42023466865) and adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for network meta-analysis [30].

2.2. Study selection

The Participant, Intervention, Control, Outcomes, and Study Design (PICOS) framework was used for study selection (Table S2). Only randomized controlled trials (RCTs) with a parallel or cross-over design were included, while cluster and multi-arm RCTs were excluded. For eligible cross-over trials, only phase 1 data were analyzed to avoid residual effects. Participants were adults aged 18 years or older with blood pressure values of 120–139/80–89 mmHg who were not receiving antihypertensive medication according to hypertension treatment guidelines [31]. This blood pressure range-based inclusion criterion also allows for the inclusion of studies with subjects diagnosed as having stage I hypertension according to the 2017 American College of Cardiology/American Heart Association (ACC/AHA) guideline [32]. Studies published in English or Chinese were included, with exclusions for studies with unclear inclusion/exclusion criteria, subjects with normal blood pressure or high blood pressure, studies that did not report prehypertension data separately, or interventions that lasted less than one week. For studies with outcomes reported at multiple time points, the end point was defined as the end of the last intervention period. If both end-of-intervention and follow-up outcomes were available, only the end-of-intervention outcomes were included in the network meta-analysis.

2.3. Data extraction

Two researchers (S.J. Cai and C.H. Wu) independently extracted data using a standardized form. The information extracted included study identifiers, location of clinical study sites, sample size, lifestyle intervention groups, duration of intervention, and outcomes. Any discrepancies were resolved through discussion or verified by a third researcher

(Y.C. Yang) to reach a consensus. The primary outcome variables of interest were SBP and DBP measured at the end point of the included trials.

2.4. Bias assessment

The risk of bias in the included studies was assessed independently by two researchers (S.J. Cai and C.H. Wu) using the revised Cochrane Risk of Bias tool (ROB 2) for randomized trials [33]. The ROB 2 checklist includes the following domains: randomization process, deviations from intended interventions, missing outcome data, measurement of the outcome, and selection of the reported results. The included studies were categorized as having a "high risk of bias", "some concerns", or "low risk of bias". The overall bias was determined by calculating the algorithm's overall judgment using the ROB 2 tool.

2.5. Lifestyle intervention definition

In this study, lifestyle interventions were categorized based on the specific behaviors participants adopted, either individually or in combination. Specifically, *Control (CO)* was defined as participants who did not receive any lifestyle intervention and maintained the original habits. *Dietary supplement (DS)* was defined as participants received an intervention involving dietary extract or supplement (e.g. Hawthorn green tea [34], a combination supplement of polyphenol extract and L-citrulline [35], black sesame meal capsules [36], red ginseng [37], c12 Peptide [38], ethanolic extract of whole tomato fruit [39], garlic preparation [40], lavender extract [41], grape seed extract [42], safflower seed extract [43], soursop supplementation [44], fish oil [45], commercially available combination dietary supplement capsules [46], water-soluble vitamins [47], Khar-i-khasak [48], Hwangchil–Unripe Bokbunja extract mixture [49], Aronia berry (poly)phenol supplementation [50], hibiscus sabdariffa [51], taurine supplementation [52], Nisha Amalaki Churna [53]). *Exercise (EX)* was defined as participants engaged in any form of exercise intervention (e.g., Baduanjin [54], aerobics exercise [55,56], yoga [57,58], Mat Pilates training [59], combined aerobic and resistance training [60], resistance band exercise training [61], arm swing exercise [62,63], lifestyle physical activity [64], isometric handgrip exercise [65], isometric exercise training [66], tai chi [67], endurance exercise training [68]). *Health education (HE)*was defined as participants received general health advice, including dietary guidance and exercise recommendations, without a specific intervention. *General healthy diet (GHD)* was defined as participants received a specific dietary intervention aimed at lowering blood pressure, not just dietary guidance, unless it is a specific dietary pattern, such as *DASH* (abbreviation of Dietary Approaches to Stop Hypertension). *Psychological care (PC)* was defined as participants received mental health care and management. *Meditation (ME), Relaxation therapy (RT)*, and *Massage (MA)* were defined according to their original definitions. *TEAS* was an abbreviation for transcutaneous electrical stimulation of acupuncture points. *Lifestyle modification* (*LSM*) was defined as a combination of interventions in addition to those described above, including blood pressure monitoring, alcohol restriction, quitting smoking, etc.

2.6. Statistical analysis

All statistical analyses were performed using RStudio (version 4.4.0). Intervention networks were established by studies that linked three or more interventions through direct comparisons. Network meta-analysis allows both direct and indirect comparisons of interventions within these intervention networks, using a common comparator as the control defined in this meta-analysis [69–71]. A Bayesian network meta-analysis was conducted using Markov Chain Monte Carlo methods, using the *"BUGSnet"* package [72].

The model selection process involved generating leverage value plots for fixed-effect and random-effect models using the *nma.fit()* function. These plots display the corresponding effective number of parameters, the total residual deviance, and the deviance information criterion (DIC). The DIC is a standard criterion for Bayesian model selection and comparison [73]. Based on the DIC values and leverage plots, the fixed-effect model was chosen when the DIC difference between the two models was $<$ five; if the DIC difference was $\geq$ five, the random-effect model with a smaller DIC was selected. To test for inconsistent assumptions, the parameter *type* = *"inconsistency"* was applied to the *nma.model()* function.

The mean difference (MD) was calculated as the effect size using the mean and standard deviation (SD) of SBP and DBP between before and after intervention reported by each included study. In cases where only the standard error was reported, we applied standard deviation transformations using the formula recommended in the Cochrane Handbook [74].

To visualize the network structure of all lifestyle interventions, a network plot was generated for each outcome of interest. Forest plots were used to visually compare the effect sizes of lifestyle interventions relative to the control group, expressed as MD with 95 % credible interval (CrI). A heat plot visualized all relative effect estimates from the network meta-analysis, including potential comparisons. Ranking probabilities for intervention efficacy were displayed using rankograms and surface under the cumulative ranking curve (SUCRA) plots, offering insights into the relative effectiveness of each intervention for pre-hypertension management.

Furthermore, both funnel plot and Egger's test were performed to assess publication bias, heterogeneity, and asymmetry. If publication bias was detected, sensitivity analysis was conducted to test the robustness of the data by excluding studies with a high risk of bias. The degree of between-study heterogeneity was examined using the size of the I-square statistic (I^2). I^2 values of 25 %, 50 %, and 75 %, respectively, indicate low, medium, and high heterogeneity, respectively [75]. If I^2 exceeds 75 %, the source of heterogeneity was investigated with *data.plot()* function, and subgroup analyses were performed to account for the identified sources of heterogeneity.

3. Result

3.1. Study selection and characteristics

Our search identified 31,483 references, which were reduced to 25,164 studies after removing duplicates. After title and abstract screening, 24,487 studies were excluded, and a further 610 studies were removed after a thorough full-text review. Consequently, this network meta-analysis included 67 eligible studies (Fig. 1) [34–68,76–107], which compared the effects of 17 types of lifestyle interventions with the control group. The 17 lifestyle interventions analyzed in this study included five single-component interventions, seven two-component combined interventions, three-component combined interventions, and one four-component combined intervention.

The baseline characteristics of the included studies are summarized in Table S3. After bias assessment, 10 studies were considered to be at high risk of bias, 43 studies were considered to be of some concern, and 14 studies were considered to be at low risk of bias (Figs. 2 and S1). Articles rated as high risk of bias were mainly due to the lack of blinding in the randomization process and bias in outcome measures.

3.2. Network meta-analysis for SBP outcome

The results of the network meta-analysis for the SBP outcome are shown in Fig. 3, with network characteristics outlined in Table S4. Leverage plots shown in Fig. S2 were used to evaluate model fit, and the random-effects model was considered more appropriate for the SBP outcome based on the DIC value (DIC = 261.00, Dres = 136.93). Furthermore, the fit index, as shown in Fig. S3, confirmed that the consistency assumption was met in our analysis.

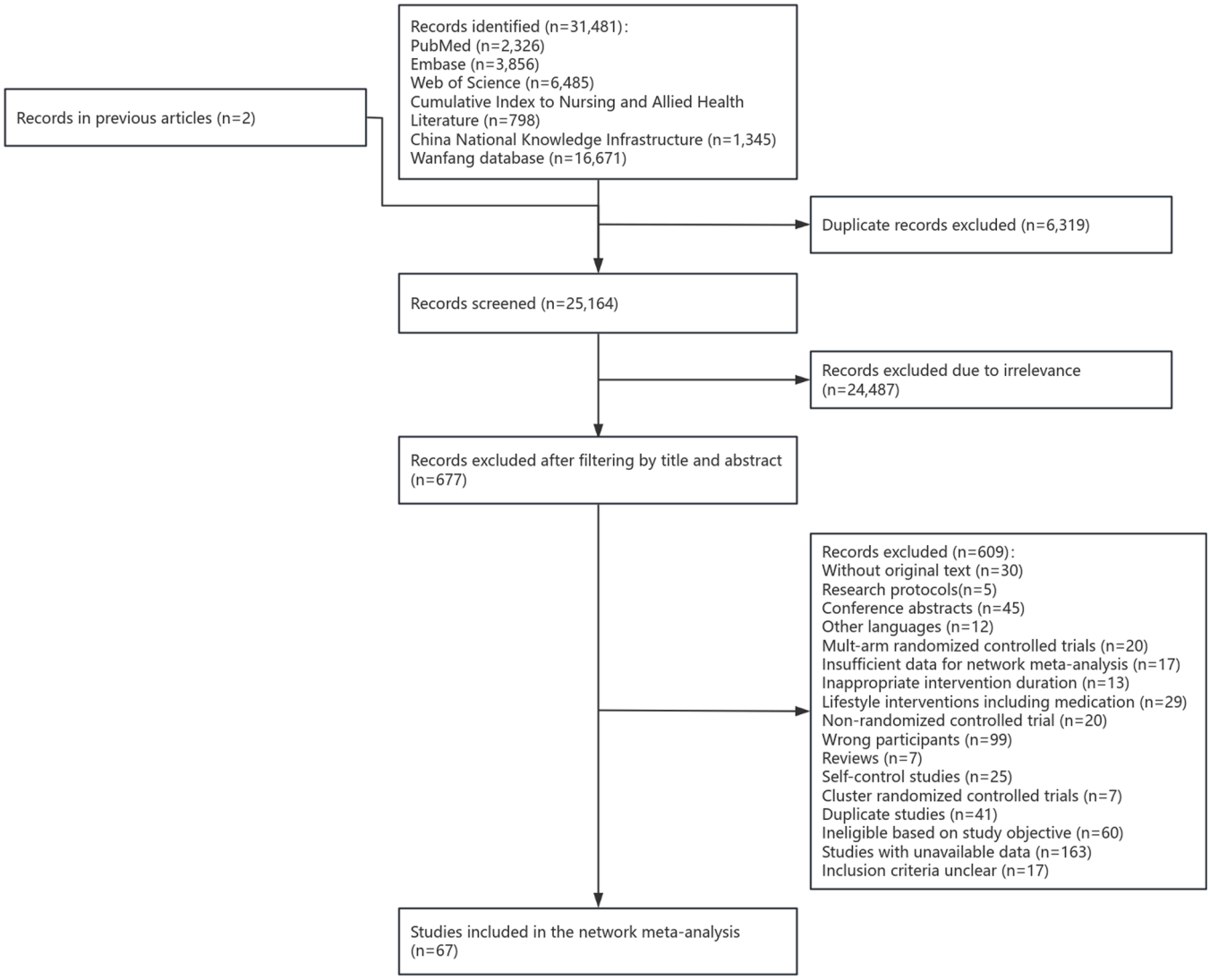


Fig. 1. Flowchart for searching and selecting eligible studies.

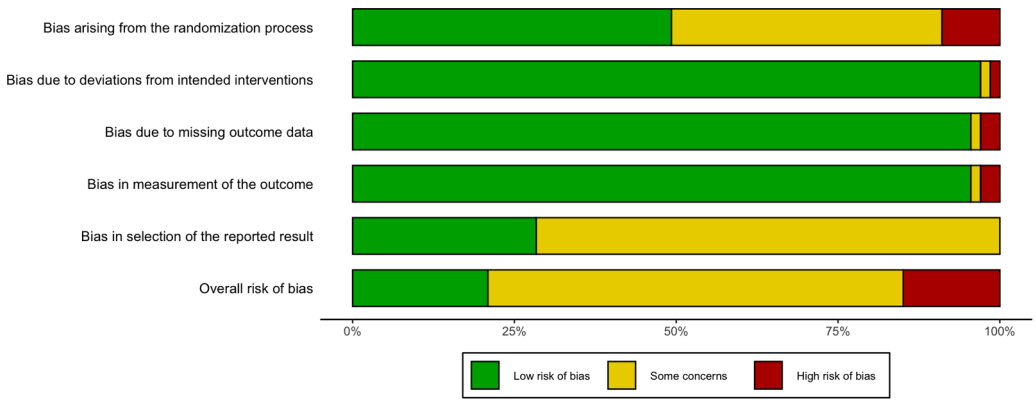


Fig. 2. Risk of bias assessment as a percentage.

Fig. 3A presents the effects of different lifestyle interventions on SBP in prehypertensive populations. The network meta-analysis included 17 lifestyle interventions and *CO*. Compared with *CO, DS plus EX* was the most effective lifestyle intervention for reducing SBP (MD −14.86, 95 % CrI −20.44 to −9.35) (Figs. 3B and S4A). Other lifestyle interventions also showed significant reductions in SBP, including *DS* (MD −3.85, 95 % CrI −5.91 to −1.84), *DS plus HE* (MD −7.76, 95 % CrI −13.26 to −2.14), EX (MD −6.17, 95 % CrI −8.96 to −3.39), *HE plus EX* (MD −10.84, 95 % CrI −17.59 to −4.04), *HE plus GHD* (MD −9.23, 95 % CrI −16.19 to −2.00), *HE plus LSM plus DASH* (MD −14.52, 95 % CrI −20.46 to −8.58), *HE plus LSM plus EX* (MD −8.32, 95 % CrI −14.34 to −2.37), *HE plus LSM plus GHD* (MD −7.98, 95 % CrI −12.90 to −3.00), *HE plus*

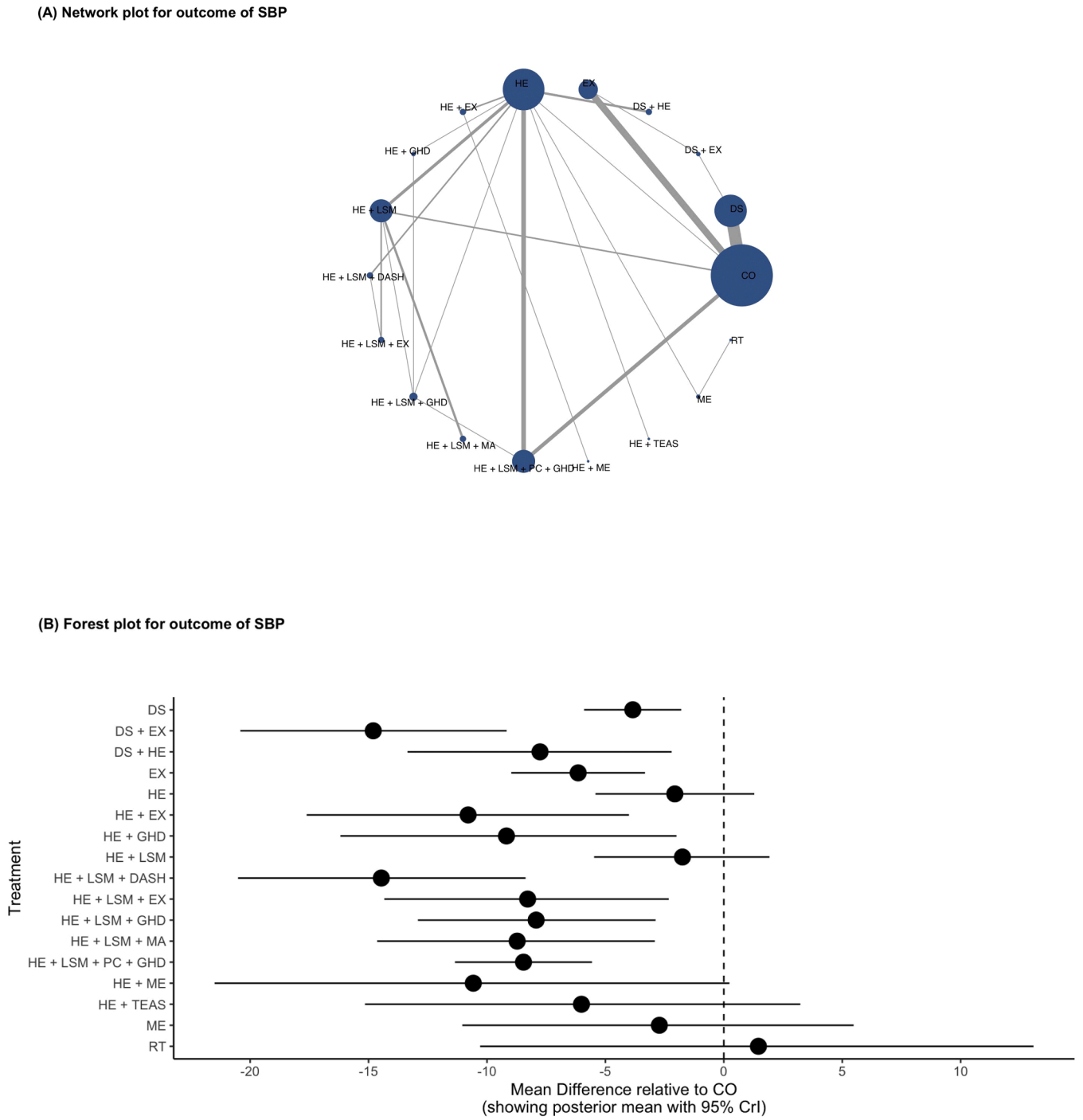


Fig. 3. Network plot and forest plot of SBP outcome. (A) network plot, each node corresponds to a specific lifestyle intervention, with "CO" denoting the control intervention. the nodes represent lifestyle interventions, the node size represents the number of studies, the lines represent the number of direct comparisons between lifestyle interventions. (B) forest plot was used to compare the effect sizes of lifestyle interventions relative to the control group. Abbreviations: SBP, systolic blood pressure; DS, dietary supplement; EX, exercise; HE, health education; LSM, lifestyle modification; GHD, general healthy diet; DASH, dietary approaches to stop hypertension; MA, massage; PC, psychological care; ME, meditation; RT, relaxation therapy; TEAS, transcutaneous electrical stimulation of acupuncture points; CO, control; CrI, credible interval.

LSM plus MA (MD −8.77, 95 % CrI −14.65 to −3.04), *HE plus LSM plus PC plus GHD* (MD −8.45, 95 % CrI −11.29 to −5.57). The rankogram plot (Fig. S5A) and the SUCRA plot (Fig. S6A) showed that the SUCRA value of reducing SBP of *DS plus EX* was 92.39 %.

We observed considerable heterogeneity among the included studies, with an I^2 statistic of 97.3 %. This variability is unlikely to be due to publication bias, as confirmed by the non-significant p-value (p = 0.071). This finding is further supported by the funnel plot (Fig. S7A). Duration of intervention, age, and economic level of the region were identified as sources of heterogeneity (Fig. S8). Subgroup analyses were then performed to explore the impact of these sources of heterogeneity.

3.3. Network meta-analysis for DBP outcome

The network characteristics for the DBP outcome are presented in Table S4. Leverage plots, shown in Fig. S2, illustrate the effect of model

fit on the analysis. According to the DIC value, we found that the random-effects model fit was better for DBP outcomes (DIC=255.69, Dres=132.40). The fit index obtained from the model indicates that our analysis meets the consistency assumption (Fig. S3).

Fig. 4A shows the effects of different lifestyle interventions on DBP in prehypertensive populations. Compared with *CO, HE plus ME* was the most effective lifestyle intervention (MD −13.99, 95 % CrI −22.25 to −5.65) (Figs. 4B and S4B). Other lifestyle interventions also showed potential to reduce DBP, including *DS* (MD −2.53, 95 % CrI −4.00 to −1.07), *DS plus EX* (MD −7.23, 95 % CrI −11.22 to −3.34), *DS plus HE* (MD −4.69, 95 % CrI −8.55 to −0.75), *EX* (MD −3.67, 95 % CrI −5.71 to −1.67), *HE plus EX* (MD −5.19, 95 % CrI −9.82 to −0.44), *HE plus LSM* (MD −3.86, 95 % CrI −6.39 to −1.26), *HE plus LSM plus DASH* (MD −10.47, 95 % CrI −14.70 to −6.30), *HE plus LSM plus EX* (MD −7.28, 95 % CrI −11.43 to −3.13), *HE plus LSM plus MA* (MD −8.48, 95 % CrI −12.50 to −4.43), *HE plus LSM plus PC plus GHD* (MD −7.42, 95 % CrI

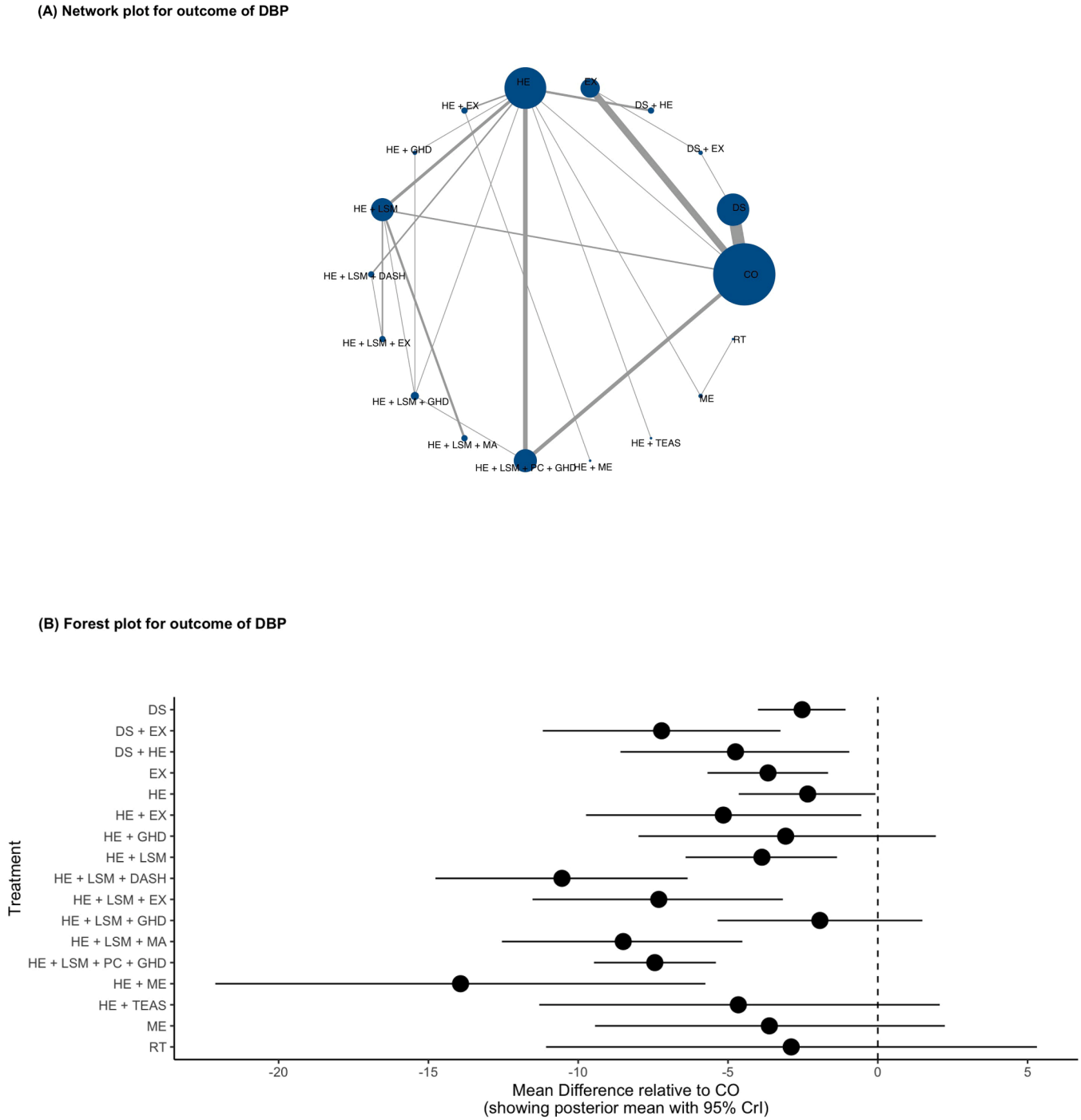


Fig. 4. Network plot and forest plot of DBP outcome. (A) network plot, each node corresponds to a specific lifestyle intervention, with "CO" denoting the control intervention. the nodes represent lifestyle interventions, the node size represents the number of studies, the lines represent the number of direct comparisons between lifestyle interventions. (B) forest plot was used to compare the effect sizes of lifestyle interventions relative to the control group. Abbreviations: DBP, diastolic blood pressure; DS, dietary supplement; EX, exercise; HE, health education; LSM, lifestyle modification; GHD, general healthy diet; DASH, dietary approaches to stop hypertension; MA, massage; PC, psychological care; ME, meditation; RT, relaxation therapy; TEAS, transcutaneous electrical stimulation of acupuncture points; CO, control; CrI, credible interval.

−9.47 to −5.40). Among them, *HE plus ME* had the highest probability of reducing DBP (SUCRA: 99.76 %) based on the rankogram and SUCRA plot (Figs. S5B and S6B).

We identified considerable heterogeneity among the included studies ($I^2 = 97.3$ %), with heterogeneity detected to be associated with the duration of intervention, age, and economic level of the region (Fig. S8). Additionally, we detected evidence of publication bias ($p < 0.001$) (Fig. S7). We therefore performed sensitivity analyses excluding studies at high risk of bias, and the results of these sensitivity analyses were consistent with those of the original network meta-analysis (Fig. S9).

3.4. Subgroup analyses

Given the observed heterogeneity among the included studies, we conducted subgroup analyses to examine the potential effects of intervention duration, age, and economic level of the region.

3.5. Subgroup analysis for SBP by intervention duration

Our analysis revealed a variation in the types of lifestyle interventions and their effectiveness based on the duration of the intervention. For lifestyle interventions lasting over six months (including 22 studies), the most effective intervention for reducing SBP lasting over six months was *HE plus LSM plus DASH* (MD −17.20, 95 % CrI −24.10 to −10.23) (Fig. 5A). In contrast, for interventions lasting less than six months (including 45 studies), *DS plus EX* was the most effective lifestyle intervention for reducing SBP (MD −15.29, 95 % CrI −21.00 to −9.68) (Fig. 5B). Furthermore, the efficacy of the same lifestyle intervention in reducing SBP appeared to be influenced by the duration. For example, the efficacy of *EX* on SBP was statistically significant for interventions lasting less than six months, whereas no significant effect was observed for interventions lasting over six months.

3.6. Subgroup analysis for SBP by age

Fig. 5C and D show the effect of lifestyle interventions on SBP in participants of different age groups. In participants aged 60 years or older (including 10 studies), *DS plus EX* was the most effective lifestyle intervention for reducing SBP (MD −16.56, 95 % CrI −32.03 to −1.46). Additionally, *HE plus LSM plus PC plus GHD* (MD −8.03, 95 % CrI −14.99 to −0.95) was also statistically significant for reducing SBP (Fig. 5C). In participants younger than 60 years of age (including 56 studies), *HE plus LSM plus DASH* was the most effective lifestyle intervention for reducing SBP (MD −14.32, 95 % CrI −21.34 to −7.19) (Fig. 5D).

3.7. Subgroup analysis for SBP by the economic level of the region

Based on the World Bank classification, we categorized the regions included in the studies into high-income countries and LMICs [108]. Of these, 18 studies were conducted in high-income countries and 49 studies were conducted in low-income countries. In high-income countries, there was no statistically significant difference in the effect of all lifestyle interventions on SBP compared with *CO* (Fig. 5E). However, in LMICs, *DS plus EX* was the most effective in reducing SBP (MD −17.48, 95 % CrI −23.31 to −11.71) (Fig. 5F).

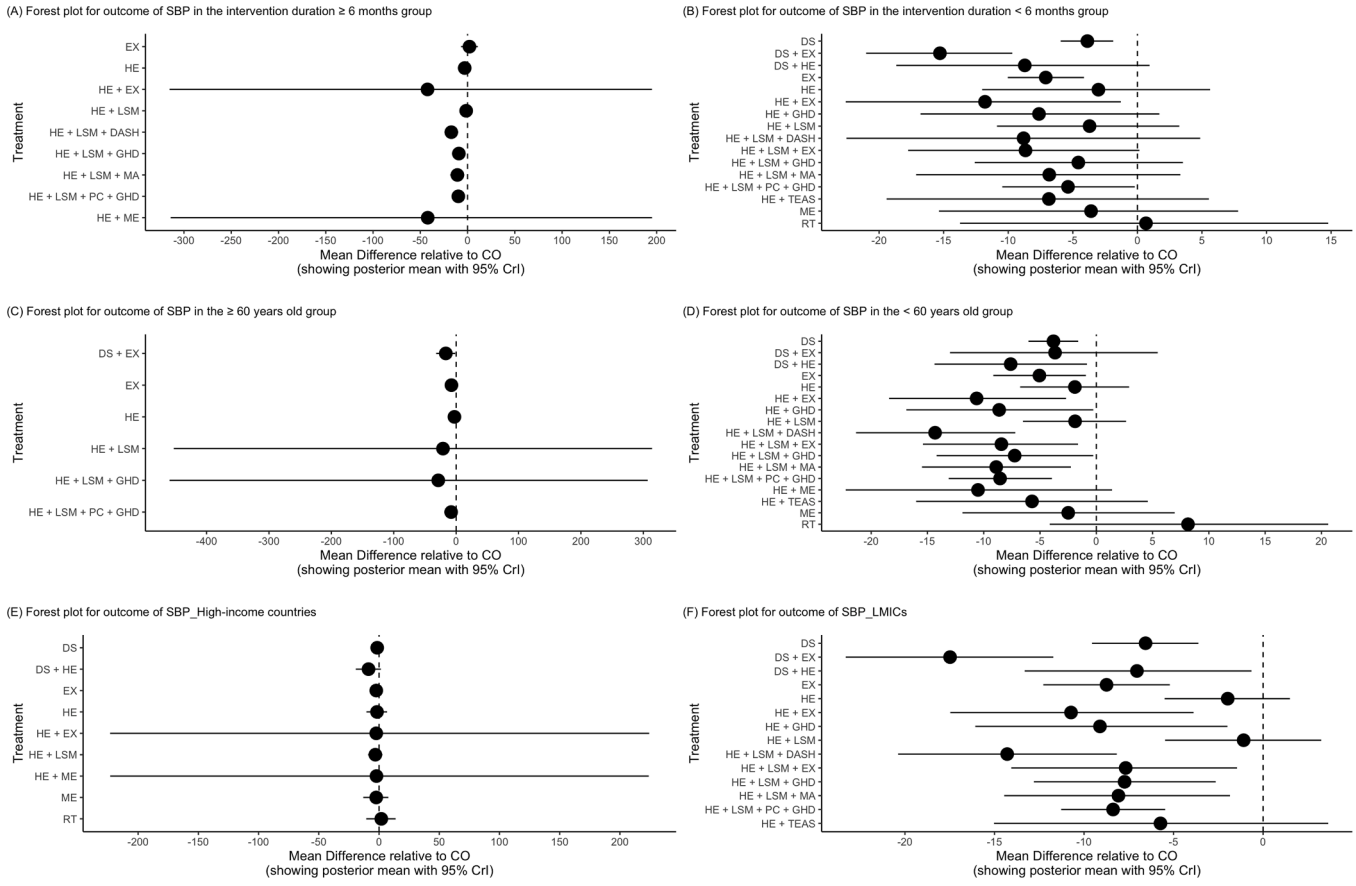


Fig. 5. Subgroup analyses of SBP outcome. Abbreviations: SBP, systolic blood pressure; DS, dietary supplement; EX, exercise; HE, health education; LSM, lifestyle modification; GHD, general healthy diet; DASH, dietary approaches to stop hypertension; MA, massage; PC, psychological care; ME, meditation; RT, relaxation therapy; TEAS, transcutaneous electrical stimulation of acupuncture points; CO, control; CrI, credible interval; LMICs, low- and middle-income countries.

3.8. Subgroup analysis for DBP by intervention duration

For interventions lasting over six months, *HE plus LSM plus DASH* (MD −10.87, 95 % CrI −17.33 to −4.31) was the most effective lifestyle intervention for reducing DBP. Additionally, *HE plus LSM plus MA* (MD −7.07, 95 % CrI −13.81 to −0.29), *HE plus LSM plus PC plus GHD* (MD −6.94, 95 % CrI −10.40 to −3.42) were also effective (Fig. 6A). For interventions of less than six months, *HE plus LSM plus MA* (MD −11.53, 95 % CrI −17.95 to −5.18) was the most effective lifestyle intervention (Fig. 6B).

3.9. Subgroup analysis for DBP by age

For people aged 60 years or older, the most effective intervention for DBP reduction was *HE plus LSM plus PC plus GHD* (MD −6.23, 95 % CrI −7.04 to −5.42). Other interventions that significantly reduced DBP included *DS plus EX* (MD −5.03, 95 % CrI −7.48 to −2.56), *EX* (MD −1.96, 95 % CrI −3.68 to −0.25), and *HE* (MD −3.44, 95 % CrI −4.49 to −2.38) (Fig. 6C). *HE plus ME* (MD −13.76, 95 % CrI −22.08 to −5.45) was shown the most significant for reducing DBP in people younger than 60 years (Fig. 6D).

3.10. Subgroup analysis for DBP by the economic level of the region

For the DBP outcome, we also identified nine distinct types of lifestyle interventions conducted in high-income countries (18 studies). In high-income countries, *EX* (MD −3.83, 95 % CI −6.85 to −1.02) was statistically significant for reducing DBP (Fig. 6E). In contrast, HE *plus LSM plus DASH* was the most effective lifestyle intervention for reducing DBP in LMICs (MD −10.82, 95 % CI −15.53 to −6.13) (Fig. 6F).

4. Discussion

This network meta-analysis evaluated the comparative effectiveness of different lifestyle interventions on SBP and DBP in people with prehypertension. A total of 67 RCTs assessing 17 different interventions and control were included. Our findings showed that *DS plus EX* was the most effective lifestyle intervention for reducing SBP, and *HE plus ME* was the most effective way for reducing DBP. Additionally, multi-component lifestyle interventions showed superior efficacy in managing blood pressure compared with single-component interventions (e.g., *DS, EX*). Notably, this study revealed heterogeneity in the blood pressure measurement outcomes across different intervention durations, age groups, and regions with different economic levels.

Dietary supplements such as omega-3 fatty acids [109,110], nitrate [111], polyphenols [112], and certain vitamins [113] have been shown to support cardiovascular health and lower blood pressure. Our analysis supports their use in the management of prehypertension. The mechanisms may be attributed to improving endothelial function, reducing oxidative stress, and regulating vascular tone [109,110,114]. Given their ease of use and strong evidence, dietary supplements could be a practical strategy for the prevention of prehypertension. However, it is crucial to assess the efficacy of different types of supplements individually, as responses may vary in different populations. Beyond individual nutrients, overall dietary patterns such as the DASH diet-which emphasizes fruits, vegetables, low-fat dairy products, fish, nuts, legumes, and reduced sodium intake-have been consistently shown to lower blood pressure [115–117]. Previous systematic reviews have reported that

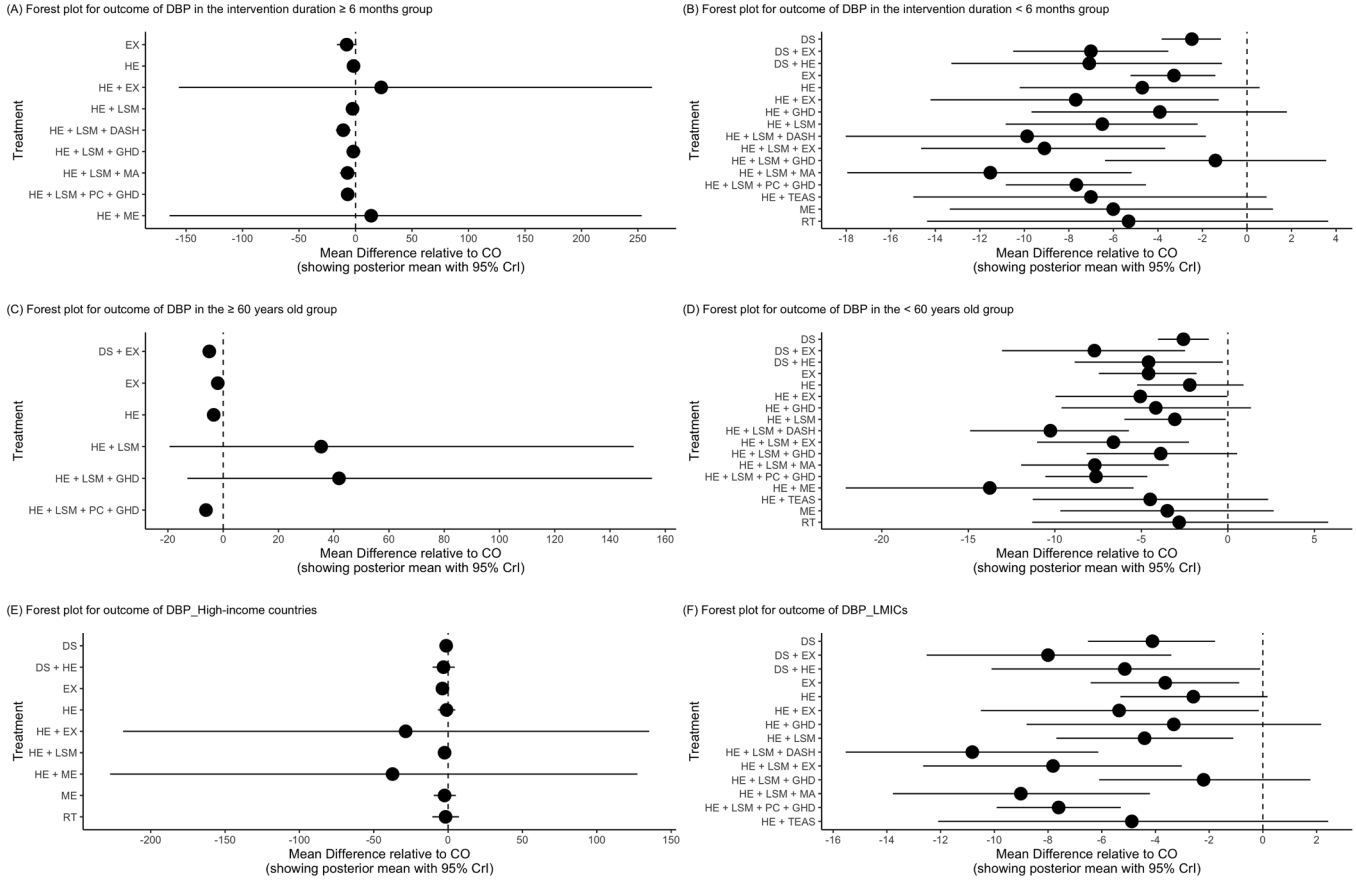


Fig. 6. Subgroup analyses of DBP outcome. Abbreviations: DBP, diastolic blood pressure; DS, dietary supplement; EX, exercise; HE, health education; LSM, lifestyle modification; GHD, general healthy diet; DASH, dietary approaches to stop hypertension; MA, massage; PC, psychological care; ME, meditation; RT, relaxation therapy; TEAS, transcutaneous electrical stimulation of acupuncture points; CO, control; CrI, credible interval; LMICs, low- and middle-income countries.

both the DASH diet and the modified DASH diet can lower blood pressure levels in adults with hypertension and prehypertension [24,118]. Our results support the conclusions of previous reviews.

Consistent with previous studies, our meta-analysis supports the beneficial effects of *EX* on blood pressure reduction [119–121]. However, the effectiveness of *EX* may be influenced by factors such as individual characteristics, *EX* type, and intensity. For example, a meta-analysis focusing on adults with different hypertensive states found that endurance training had the greatest reduction in blood pressure in hypertensive individuals, whereas dynamic resistance training had a modest but non-significant effect in those with prehypertension [122]. Another research found that moderate-intensity resistance training significantly reduced blood pressure, while low-intensity interventions did not find this association [123]. Due to the limited number of studies for each exercise type, we were unable to perform subgroup analyses in the prehypertensive population. Future research should focus on identifying the most effective *EX* modalities for optimal outcomes in prehypertension.

Our results suggest that combining multi-component lifestyle interventions improves blood pressure management in individuals with prehypertension. For example, *DS* reduced SBP by −3.85 mmHg, but when combined with *HE*, the reduction increased to −7.76 mmHg. A similar trend was observed for DBP, suggesting that integrating lifestyle changes with dietary factors offers greater benefits. Specifically, the combination of diets such as *DASH* or *GHD* with other interventions appeared to be more effective than these diets alone, likely due to the greater weight loss achieved with exercise and a healthy diet, together with strong adherence to lifestyle changes [124].

A previous network meta-analysis showed that comprehensive lifestyle interventions are superior to exercise alone in reducing blood pressure in children and adolescents [125]. Our study extends this finding to adults, further reinforcing the existing evidence. One possible explanation is that multiple lifestyle interventions may be more effective than addressing individual behaviors in isolation, because achieving one lifestyle change goal may create additional opportunities to achieve other goals [126,127]. These findings support current guidelines, which suggest that multi-component interventions are more likely to be effective when implemented in real-world settings [128]. However, more research is needed to explore the effects of combining exercise with other lifestyle interventions.

Another significant finding of our study is that *HE* alone did not significantly reduce blood pressure in the general population. This may be because it does not bridge the gap between knowledge and action, thereby limiting its effectiveness as an intervention [129]. While health education may affect the cognitive aspect, it can enhance an individual's understanding, motivation, and adherence to recommended lifestyle changes [130]. This highlights the need to combine education with behavioural interventions to improve adherence and outcomes.

ME as a mind-body intervention has been shown to be beneficial in lowering blood pressure by reducing symptoms of depression, improving emotional stability, and regulating vascular health [131–134]. In our study, we found that *HE plus ME* was an effective approach to lowering DBP, but in contrast to the previous meta-analysis study, we did not observe a significant effect of *ME* on SBP [135]. This discrepancy may stem from the different underlying physiological mechanisms of SBP and DBP. SBP, which reflects the pressure during heartbeats, is mainly influenced by cardiac output, circulating blood volume, and aortic elasticity [136,137]. In contrast, DBP, which measures arterial pressure during myocardial relaxation, is more closely related to peripheral vascular resistance and the elastic properties of large arteries [136,137]. *ME* may exert a more pronounced effect on DBP by modulating the sympathetic-parasympathetic nervous system balance, thereby reducing peripheral vascular resistance and overall vascular tone [138,139]. Given the limited number of included *ME* studies, further research with larger samples is needed to clarify the mechanisms.

Our study conducted stratified analyses to explore how different interventions affect blood pressure in different populations. We observed that *HE plus LSM plus DASH* was the most effective way to reduce SBP and DBP in interventions lasting over six months, probably due to the cumulative benefits of dietary changes on metabolic and vascular health. In shorter interventions, *DS plus EX* had the greatest effect on SBP, possibly due to rapid improvements in vascular tone and elasticity, which directly lower blood pressure [113,119,140]. *HE plus LSM plus MA* was most effective in reducing DBP in shorter interventions, probably due to the mechanical and thermal effects of massage that improve arterial circulation [141].

The duration of the intervention also seemed to influence the effectiveness of lifestyle changes in lowering SBP. Our results, based on a limited number of long-term interventions, contrast with previous studies suggesting that long-term *EX* interventions are more effective [142]. This highlights the importance of early initiation and sustained adherence to *EX*. However, poor adherence to *EX* may pose a challenge to long-term interventions and achieving effective results. To address this issue, improving adherence to exercise can be combined with behavioural strategies such as self-monitoring, goal setting, and feedback to develop a comprehensive strategy [143,144].

Subgroup analyses also showed age-related differences in the effectiveness of lifestyle interventions. In participants aged 60 years and older, the most effective interventions for reducing SBP and DBP were *DS plus EX* and *HE plus LSM plus PC plus GHD,* respectively. This supports previous research highlighting the benefits of resistance training and dietary supplements for older adults and the value of integrated health management for mental and physical health [145,146]. In addition, given the two-way relationship between hypertension and mental health problems, older people with prehypertension may benefit from integrated health management that includes psychological care [147]. In contrast, for participants younger than 60 years, *HE plus LSM plus DASH* was most effective for reducing SBP, while *HE plus ME* was most effective for reducing DBP. This suggests that younger individuals may be more responsive to comprehensive *HE* and behavioural interventions that encourage long-term lifestyle changes. These observations highlight the importance of tailoring intervention strategies to the specific needs and characteristics of different age groups.

Our results are consistent with a systematic review demonstrating the effectiveness of lifestyle interventions in lowering blood pressure in LMICs [142]. In contrast, our study found no statistically significant effect of lifestyle interventions on SBP in high-income countries. This may be attributed to the relatively limited number of studies conducted in high-income countries (only 18). Additionally, since the 1980s and 1990s, high-income countries have made significant advances in recognizing, treating, and controlling hypertension [148], which may have diminished the relative impact of these simple lifestyle interventions. Furthermore, social determinants of health may also play a key role in blood pressure control in high-income countries [149,150]. However, our study did not explicitly address these factors. These findings highlight the need to consider economic development, social determinants, and population-specific health needs when designing lifestyle interventions. Future studies should include more diverse and representative populations.

4.1. Strengths and limitations

This network meta-analysis has several strengths. Firstly, it focuses specifically on individuals with prehypertension, offering actionable insights into the management of this group. Secondly, it evaluates a wide range of lifestyle interventions, both individually and in combination, capturing the interactions between different strategies and their impact on blood pressure control. Thirdly, we addressed heterogeneity by conducting stratified analyses across key subgroups (e.g., duration of intervention, age, and economic level of the region), allowing for a more tailored approach to prehypertension. However, there are some

limitations. The heterogeneity of the included studies may have influenced the results, despite stratified analyses. While the risk of bias was assessed using the ROB 2 tool and sensitivity analyses were performed, bias remains a potential issue. Due to differences in the measurement and reporting of lifestyle intervention measures, this study was unable to clearly define the dose or threshold for their effectiveness. This limitation restricts our ability to determine the optimal design for various lifestyles. Additionally, the majority of the studies were from LMICs, which limits the generalisability of the findings to high-income settings. Future research should include more diverse populations to increase external validity and ensure broader applicability in different economic contexts.

5. Conclusion

This network meta-analysis evaluated the effectiveness of different lifestyle interventions in the management of prehypertension. Our results suggest that *DS plus EX* was the most effective lifestyle intervention for reducing SBP in individuals with prehypertension, and *HE plus ME* was the most effective lifestyle intervention for reducing DBP. This study underlines the importance of dietary factors and exercise in the management of prehypertension and highlights the superiority of multicomponent lifestyle interventions. Furthermore, tailored approaches are essential to address the increasing burden of prehypertension in different populations.

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Ethical approval

No ethical approval was required for the study.

CRediT authorship contribution statement

Yucheng Yang: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiaona Na:** Data curation, Conceptualization. **Sijia Cai:** Investigation, Data curation. **Chuhan Wu:** Investigation, Data curation. **Shuo Han:** Writing – review & editing. **Kai Yu:** Writing – review & editing, Supervision. **Ai Zhao:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ajpc.2025.101045.

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